

Order imbalance, market returns and macroeconomic news

Evidence from the Australian interest rate futures market

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1. Introduction

The relationship between trading activity and the returns of financial assets has been examined by an array of literature. Many early studies measure trading activity by volume. Foster and Viswanthan (1990), Hiemstra and Jones (1994) and Lo and Wang (2000) study the US equity market and find that volume is positively related to returns, and closely linked to liquidity. However, measuring trading activity by volume may create problems and may actually conceal information. Trading volume can be high either due to a preponderance of buyer-initiated or seller-initiated trades, or because there is a large amount of trading interest on a given day, which is evenly distributed between buyers and sellers, each possibility having implications for prices and liquidity.

Order imbalance, defined as the difference between buyer-initiated and seller-initiated trades, is a measure of trading activity that has been suggested as been more informative than volume. The impact of order imbalance on returns and liquidity may be the result of information asymmetry, or inventory adjustment. Glosten and Milgrom (1985) and Kyle (1985) develop theoretical models which assume that the trades of market participants will reveal information to the market when they have private information about the value of an asset. In equilibrium, the sensitivity of prices, and prevailing liquidity, will depend on the level of information asymmetry. The inventory models of Stoll (1978) and Ho and Stoll (1983) provide an alternative explanation; a large order imbalance may exacerbate the inventory problem faced by market-makers who will respond by changing bid-ask spreads and amending price quotations. More extreme order imbalances should have a greater effect on prices and liquidity owing to the possibility of asymmetric information, and inventory adjustment problems.

A number of articles have considered order imbalances around specific events; Blume et al (1989) study order flow around the October 1987 crash and find a strong relation between order imbalances and stock price movements together with evidence of subsequent reversals, whilst Lee (1992) considers earnings announcements and find that good news triggers brief but intense buying pressure. Chan et al (1999), Chan and Fong (2000), and Hasbrouck and Seppi (2001), study order imbalance in US equity markets over relatively short periods and find that there is a strong predictive ability for subsequent stock returns. Chordia et al (2002) conducted the first extended study using order imbalance on NYSE stocks and found that order imbalances are strongly related

to contemporaneous absolute returns, as well as past market returns, and that investors exhibit contrarian behaviour in aggregate.

Order imbalance methodology has also been applied to the investigation of financial market reaction to macroeconomic data announcements. Evans and Lyons (2002) show that foreign exchange order flow predicts macroeconomic surprises. Green (2004) develops a structural model to examine the informational content of trading in the US Treasury market surrounding US macroeconomic announcements. Sensitivity of prices to order flow is lower than usual before announcements, which is consistent with no information leakage; following the announcements there is an increased sensitivity to order flow, suggesting the release of public information increases the level of information asymmetry. Pasquariello and Vega (2007) employ a parsimonious model of speculative trading to analyse the response of two-year, five-year, and ten-year US bond prices to order flow and macroeconomic news over the period 1992-2000 and find that unanticipated order flow has a significant impact on daily bond price changes, this effect is greater when the dispersion of beliefs among market participants is high. Brandt and Kavajecz (2004) and Brand et al (2006) examine price discovery in the US Treasury market, finding that order-flow drives price movements, accounting for up to 26% of the variation in yields on days without macroeconomic announcements. Green (2004) considers the impact of trading on the prices of five-year US Treasury notes around scheduled macroeconomic releases and finds a significant increase in the informational role of trading following economic announcements. This suggests the release of public information increases the level of information asymmetry; the effect is greater after announcements with a bigger surprise component and thus a larger initial price impact. Underwood (2009) looks at the cross-market relationship between equities and bonds and notes that aggregate order imbalances play a strong role in explaining returns.

The order imbalance methodology has received little application in the Australian financial markets, and in particular within the Fixed Income space. With the increasing size, and relevance, of the Australian financial markets there is need for additional research in this area. This paper has several key aims. Firstly, a variant of the Chordia et al (2002) methodology is applied to Fixed Income markets for the first time in order to examine the relationship between order imbalance and contemporaneous price movements in the Australian Interest Rate futures markets. This analysis will also enable a deeper understanding of aggregate trading behaviour in

Interest Rate futures markets. Secondly, the link between the absolute level of order imbalance and market liquidity will be examined. This is of particular importance to market participants who require a high level of liquidity in order to successfully execute trading strategies. Finally, there is an examination of the relationship between order flow and macroeconomic announcements. This is the first such study on a non-US market and as such contributes to the literature by seeking to confirm or refute whether the existing studies are applicable only to the US, or are more broadly applicable to international markets.

The key findings are summarized as follows. Order imbalances are strongly related to contemporaneous returns even after considering aggregate market activity. Contemporaneous order imbalance exerts a significant impact on market returns in the expected direction i.e. excess buy (sell) orders drive up (down) prices. Order imbalances are related to past market returns and participants in the Australian Interest Rate futures market tend to be contrarian across all products following market rallies, but whereas investors in bond futures continue to sell following market declines, bank-bill investors remain contrarian. Nine major macroeconomic announcements are identified, and both order imbalance and returns react to such announcements in a manner that correctly reflects the news component. The reaction to macroeconomic news is also consistent with established research such that negative news has a greater impact than positive news. The impact of macroeconomic news on bond futures is greater than that for the shorter-term bank bill futures. Following a scheduled macroeconomic announcement there is an increase in the level of information asymmetry within the interest rate futures market, demonstrated by an increased sensitivity to order flow. Finally, the pattern of order imbalance prior to announcements suggests that there is no information leakage.

The rest of this paper is organized as follows: Section 2 discusses the nature of the data used in this paper. Section 3 outlines the methodology employed in the analysis. Section 4 provides discussion on the results of the empirical investigation. Section 5 concludes the paper.

2. Data Description

2.1 Australian Interest Rate Futures

Data on Australian Interest Rate Futures was collected from Thomson Reuters Tick History (TRTH), provided by SIRCA¹, for the period 6th January 2004 to 31st December 2010, a total of 1,795 trading days. Data is gathered for 90-day Bank Bill, and 3- and 10- year Government Bond Futures². To ensure that the study concentrates on the most liquid contracts only the nearest contract is considered. In-line with market convention, the contract under consideration is rolled on the last day of the month preceding the delivery month. Interest rate futures effectively trade around the clock on the Sydney Futures Exchange (SFE) with the day session covering 8.30am to 4.30pm, and the night session covering 5.10pm to 7.00am. To ensure the consistency of daily returns and trade data, the close is taken to be the end of the day-session (4.30pm Australian EST (GMT+10)).

<Insert Table 1 – Summary Statistics>

Each transaction is designated as either buyer-initiated or seller-initiated using the established Lee and Ready (1991) algorithm. If the trade occurs at the price equal to the offer (bid) prevailing immediately prior to the transaction then it is identified as a buy (sell) transaction. Fortunately, the difficulties that Lee and Ready (1991) identify in the US equity market, namely trades inside the spread and identification of the prevailing quote, are not present in the electronically traded Australian Interest Rate futures market.

For each trading day, and for each ten-second interval around major macroeconomic announcements, the following is computed for each of the identified futures contracts:

R_t : the return for the futures contract during interval t , defined as $R_t = \log (P_t / P_{t-1})$;

OIBTRA $_t$: the number of buyer-initiated trades less the number of seller initiated trades during interval t ;

OIBVOL $_t$: the volume of buyer-initiated futures contracts purchased less the volume of seller-initiated futures contracts sold during interval t ;

QSPR $_t$: the quoted bid-ask spread averaged across all trades during interval t ;

NUMTRANS $_t$: the total number of transactions during interval t ; and

¹ Securities Industry Research Centre of Asia-Pacific

² Reuters TIC: YBA, YTT, YTC respectively

VOL_{*t*}: the total futures contact volume during interval *t*.

<Insert table 2 - Summary statistics of order imbalance variables>

2.2 Macroeconomic Announcements

Ederington and Lee (1993) identify ‘major’ macroeconomic announcements as those with a statistically significant impact on market volatility. Dummy variables, D_{kt} , are defined, where $D_{kt}=1$ if an announcement *k* is made on day *t* and $D_{kt}=0$ otherwise. The dependent variable in the regression is price volatility in the 30-second interval following announcements, interval *j*, on day *t*. Following McNish and Wood (1992), in order to remove the impact of price fluctuations due solely to bid-ask bounce, volatility ($QTESD$) is calculated as the standard deviation of the quote mid-point through each ten-second interval:

$$QTESD_{jt} = a_{0j} + \sum_{k=1}^K a_{kj} D_{kt} + e_{jt} \quad (1.A)$$

$$QTESD_t = \sqrt{\frac{\sum_{i=1}^n (Q_i - \bar{Q})^2 t_i}{\sum_{i=1}^n t_i}} \quad (1.B)$$

Where t_i is the period for which Q_i , the quote mid-point, is alive during interval *t*. The coefficient a_{kj} is positive and significant if announcement type *k* has a significant impact on price volatility or approximately zero (or negative) if an announcement has little impact. Ederington and Lee (1993) note that as $(\pi/2)^{0.5} a_{0j}$ provides an estimate of volatility in interval *j* on non-announcement days, then the estimated volatility in interval *j* on days when *k* is announced is given by $1.2533(a_{0j} + a_{kj})$.

Using a significance level of 0.005 to identify ‘major’ announcements, 9 announcements are significant in explaining volatility; Building Approvals, Consumer Price Index, Employment, Gross Domestic Product, Private Capital Expenditure, Producer Price Index, RBA Target Rate, Retail Sales and Wage Cost Index. However, in order to facilitate analysis the sample is restricted to those announcements occurring at 11.30am (AEST), thus removing the RBA Target Rate announcement from the sample. Excluding the RBA Target Rate announcement, there are 392 major announcements during the 1,795 trading days in the sample period.

<Insert table 3>

3. Methodology

3.1 Order imbalance and market returns

Many factors may contribute to order imbalance: market returns, macroeconomic news, daily return and weekly volatility regularities (e.g. Gibbons and Hess (1981) and Chordia et al (2001)) and the reversal of temporary price pressures have all been cited as possible factors. This section first investigates whether order imbalance can be predicted using past market returns after controlling for weekly regularities and past lagged values of order imbalance. Equation 2 regresses the daily order imbalance in the number of transactions (OIBTRA) on past values of order imbalance, variables designed to capture past market moves and on day of the week dummies.

$$OIBTRA_t = \alpha_c + \sum_{i=1}^4 \beta_i OIBTRA_{t-i} + \sum_{j=1}^4 \beta_j \text{Min}(0, R_{t-j}) + \sum_{k=1}^4 \beta_k \text{Max}(0, R_{t-k}) + \sum_{l=1}^4 \beta_l \text{Day}_l + \varepsilon_t \quad (2)$$

To examine the relationship between market returns and order imbalances, a signed measure of order imbalance is required. In order to allow for a differential impact of excess buy and sell orders, order imbalance is split into positive and negative parts and included as separate regressors. Returns are regressed on contemporaneous and lagged positive and negative daily order imbalances, as well as lagged positive and negative index returns.

$$R_t = \alpha_c + \beta_{1,t} \text{Max}[0, OIBTRA_t] - \beta_{2,t} \text{Min}[0, OIBTRA_t] + \beta_{3,t} \text{Max}[0, OIBTRA_{t-1}] - \beta_{4,t} \text{Min}[0, OIBTRA_{t-1}] + \beta_{5,t} \text{Max}[0, R_{t-1}] + \beta_{6,t} \text{Min}[0, R_{t-1}] + \varepsilon_t \quad (3)$$

This paper argues that order imbalances provide information about movements in the price of bond futures in addition to that provided by aggregate volume. For example, if aggregate volume is driven by equal amounts of buying and selling activity, the impact of volume on price movements could be minimal, while if volume is driven by a large order imbalance it could have a large impact. Equation 4 seeks to disentangle the role of volume and order imbalance as drivers of returns, with the absolute value of returns acting as a proxy for volatility.

$$|R_t| = \alpha_c + \beta_{1,t} \text{Max}[0, OIBTRA_t] + \beta_{2,t} |\text{Min}[0, OIBTRA_t]| + \beta_{3,t} \text{Volume}_t + \beta_{4,t} |R_{t-1}| + \varepsilon_t \quad (4.A)$$

$$|R_{t+1}| = \alpha_c + \beta_{1,t} \text{Max}[0, OIBTRA_t] + \beta_{2,t} |\text{Min}[0, OIBTRA_t]| + \beta_{3,t} \text{Volume}_t + \beta_{4,t} |R_t| + \varepsilon_t \quad (4.B)$$

3.2 Order imbalance and macroeconomic news

In order to examine the inter-relationship between order imbalance and macroeconomic news in sufficient detail intra-day data is utilized. Following Ederington and Lee (1995), the

analysis examines 10-second intervals for twelve minutes around ‘major’ macro announcements; the window extends from 2 minutes before to 10 minutes after the announcement.

The first step is to understand how order flow adjusts in this twelve minute interval; does order flow predict the imminent announcement, is the magnitude of the surprise relevant, and following the announcement does the order flow proceed in the ‘appropriate’ direction? In this case, the appropriate direction for interest rate futures is determined by 1) the *policy anticipation hypothesis* which suggests that the reaction of market interest rates to economic news depends on how market participants expect monetary policy to change in reaction to the news, and 2) the *real activity hypothesis* implies that a stronger than expected piece of economic data may only be signalling that the economy is stronger than previously thought, thereby leading market participants to raise their expectations of real interest rates. A positive news surprise, that implies a stronger (weaker) than expected economy, should result in higher (lower) yields, lower (higher) futures prices, and an increase in the sell (buy) order imbalance. If there is information leakage prior to a scheduled announcement, a positive (negative) news event should be preceded by a negative (positive) order imbalance (more sellers than buyers). Regardless of the existence of leakage, a positive news event should be followed by a negative order imbalance and vice versa.

The news effect is regressed on order flow in equation 5. If the theory holds true, then the order flow regression coefficients (β) for equation 5.A should be negative. Following Green(2004) the larger the magnitude of surprise the greater the ensuing order imbalance.

$$OIBTRA_t = \beta_c + OIBTRA_{t-1} + \sum_{j=1}^9 \beta_{news} news_{j,t} + \varepsilon_t \quad (5.A)$$

$$OIBTRA_t = \beta_c + OIBTRA_{t-1} + \sum_{j=1}^n \beta_{+ve\ news} + ve\ news_{j,t} + \sum_{l=1}^m \beta_{-ve\ news} | -ve\ news_{l,t} | + \varepsilon_t \quad (5.B)$$

To compare coefficients on announcements surprise series with different magnitudes, the news is standardized as suggested by Balduzzi, Elton and Green (2001). In particular, for announcement type k on day t, the news surprise is defined as:

$$news_{kt} = \frac{A_{kt} - E_{kt}}{\sigma_k} \quad (6)$$

Where A_{kt} is the actual value, E_{kt} is the (Bloomberg) market survey expectation, and σ_k is the standard deviation of $(A_{kt} - E_{kt})$. An announcement surprise equal to 1.0 implies a surprise that is one standard deviation greater than zero for that announcement type. Since the $news_{kt}$ variable requires survey results as well as the actual release, any news announcement which does not have this information is removed from the sample.

Next, the paper exams the relative impact of order flow and macroeconomic news on the return and conditional volatility of interest rate futures, in the twelve minute interval around scheduled macroeconomic announcements. Smales (2011) characterizes the returns of Australian Interest Rate Futures as skewed, leptokurtic and non-normal distributions with time-varying second moments. Nelson (1991) illustrates that the exponential general autoregressive conditional heteroskedastic (EGARCH) model is well suited to modelling such financial returns. The univariate EGARCH(1,1) model is used to model specific forms of the order-flow and macroeconomic news effect.

$$R_t = \alpha_c + \alpha_{Lag}R_{t-1} + \alpha_{OIB}OIBTRA_t + \sum_{j=1}^n \alpha_{news}news_{j,t} + \alpha_{Mon}Mon_t + \varepsilon_t \quad (7.A)$$

$$\ln h_t = \beta_c + \beta_h \ln h_{t-1} + \beta_{\varepsilon 1} \frac{\varepsilon_{t-1}}{\sqrt{h_{t-1}}} + \beta_{\varepsilon 2} \frac{|\varepsilon_{t-1}|}{\sqrt{h_{t-1}}} + \beta_{OIB}OIBTRA_t + \sum_{j=1}^n \beta_{news}news_{j,t} + \beta_{Mon}Mon_t + \varepsilon_t \quad (7.B)$$

The conditional mean equation for the returns in the interest rate futures market series (R_t) is expressed as a function of past returns (R_{t-1}) contemporaneous order imbalance ($OIBTRA_t$), macroeconomic news effects ($news_t$) as well as a dummy variable for the Monday effect (Mon_t) that controls for more intense information flows following the closure of markets over a weekend. The conditional variance equation for the returns in the financial time series (h_t) is expressed as a function of the past variance (h_{t-1}), contemporaneous order imbalance ($OIBTRA_t$), macroeconomic news effects ($news_t$) and a Monday effect (Mon_t).

4. Empirical Results

4.1 Order imbalance and market returns

The time-series regression for equation 2 is reported in Table 4. Order imbalances are highly predictable based on lagged order imbalances; for at least 4 periods in the case of 90-Day Bank Bills and 3-Year bonds, and 3 periods in the case of 10-Year bonds. Table 4 emphasises key differences in the reaction of order imbalances in Bank Bill and Bond futures to changes in market prices. Participants in the Bank-Bill futures market tend to be significantly contrarian in nature

following both market rallies and market declines; selling as the market rises and buying as the market falls. On the other-hand, investors in Bond futures are contrarian in selling after market rallies (although this is not significant for 10-year Bonds), but appear to exhibit some continuation of selling following market falls. There is also a significant Tuesday irregularity for the order imbalance of all Australian Interest Rate futures; this coincides with the release of key macroeconomic data including the Consumer Price Index (CPI) and RBA target rate decisions.

<Insert Table 4>

Table 5 reports the results from the regression equation 3. Contemporaneous order imbalance, measured by OIBTRA, exerts an impact on market returns in the expected direction; positive coefficients imply that excess buy (sell) orders drive up (down) prices. This result is significant for excess buy and excess sell orders, across all three futures although the economic significance is much smaller for 10-year Bond futures. For Bank Bill futures the impact of excess buy orders is 59% greater than for excess sell orders, for 3-year futures the impact is similar for both buy and sell orders, whilst for 10-year bond futures the impact is nearly 100% greater for excess sell orders than for excess buy orders. Lagged excess buy orders do not exhibit any effect, but in all cases the effect of lagged excess sell orders is significant and consistent with contrarian behaviour. Including lagged returns produces significant results only for the lagged positive returns for Bank Bill futures, and signifies that positive returns in one period are followed by positive returns in the next. Since positive returns in interest rate futures are often driven by negative economic news, this finding is consistent with the more significant effect of negative news on US equity markets identified by Cox and Peterson (1994) and Chordia et al (2002), and in the US Treasury market suggested by Green (2004).

Panel B investigates the phenomena further by dividing the sample into periods of high-positive order-imbalance with high-return, and large-negative order-imbalance with low-returns, and determining whether returns are predictable. The only evidence that returns are predictable is in the 3-year bond. For large-negative-imbalance, large-negative-return days, both lagged order imbalance and lagged returns have predictive power, but for high-positive-imbalance, high-positive-return days only lagged order imbalance is statistically significant in terms of predictability. Volume is included as an additional regressor to ensure that the predictability of

returns is not driven by the level of unsigned trading volume; inclusion of this regressor does not materially affect the results.

<Insert Table 5>

Table 6 provides evidence on the assertion that order-imbalance is more informative than volume in explaining contemporaneous market returns, and volatility. For all futures both the excess buy orders and excess sell orders are statistically significant, although excess buy orders have a greater economic impact; a finding that is again consistent with negative economic news having a greater effect than positive news. The level of quoted spread plays a role in determining returns for 10-year bonds, with an increase in the quoted spread been indicative of higher return volatility. After controlling for order imbalance, volume has a significance suggesting that the overall level of market activity has an additional impact on contemporaneous returns.

The same variables are used to predict the returns on the following day (Equation 4.B). Consistent with previous research on equity markets, excess buy order imbalance remains a significant explanatory factor. Excess sell orders and lagged returns retain their significance for 90-Day Bank Bills and 3-year bonds. Aggregate volume disappears as a significant factor, suggesting that the persistence in volatility of returns is induced partly by persistent levels of excess order imbalance rather than overall market activity.

<Insert Table 6>

In summary, participants in the Australian Interest Rate futures market tend to be consistently contrarian in nature following market rallies, but whereas investors in bond futures continue to sell following market declines, bank-bill investors remain contrarian. There is a strong contemporaneous relationship between order imbalance and returns on interest rate futures, in the expected direction, such that excess buy (sell) orders drive up (down) prices. Considering contemporaneous returns, the level of aggregate market activity has an impact even after controlling for order imbalance, however when predicting next period volatility aggregate volume disappears as a significant factor suggesting that order imbalance is a more informative measure.

4.2 *Order imbalance and macroeconomic news*

Figure 1 depicts the dynamics of order imbalance in the 12-minute period surrounding scheduled macroeconomic announcements, whilst Table 7 considers the same dynamics over a shorter 2.5-minute sub-period. In general, there is no significant pattern of order imbalance in the two-minute period prior to macroeconomic news events, although there is some selling prior to both negative and positive news events. This is consistent with no information leakage prior to the data announcement. Following the release of macroeconomic news, order imbalance reacts in the expected way with excess buy (sell) orders following negative (positive) news in all three contracts. The magnitude of sell orders following positive news is greater than that of buy orders following negative news in the first twenty seconds, this is consistent with Lee (1992) equity market finding that good news triggers brief but intense buying in large trades. However, for the remainder of the period, the magnitude of sell orders is greater suggesting an extended reaction to negative news. For 90-Day bank bills there are excess buy orders for 80 seconds before a reversal occurs following a negative news release, compared to 50 seconds of excess sell orders before a reversal following positive news; bond futures produce similar results. Once more, this finding is consistent with negative news producing a greater impact on the price of financial assets than positive news.

<Insert Figure 1>

<Insert Table 7>

The release of scheduled major macroeconomic data has a significant impact on order imbalance. Table 8 shows that the two key economic indicators of Consumer Price Index (CPI) and the change in employment have significant impact on order imbalance across all three interest rate futures. In addition, key indicators of economic activity - Building Approvals, Gross Domestic Product (GDP) and Retail sales – have a significant impact on the shorter maturity 90-Day Bank Bills and 3-Year Government Bond futures. Whilst the Producer Price Index (PPI) inflation indicator has a significant relationship with contemporaneous order imbalance for both bond futures. Short-maturity futures react more closely to factors which have a close bearing on near-term central bank action, whilst the longer-term bond futures are affected by inflation expectations.

The impact of macroeconomic news on order imbalance correctly reflects the surprise component, with positive news events indicating economic strength, and a likely increase in real-yields following the Monetary Policy Expectations Hypothesis, and therefore a larger number of sell orders, while negative economic news indicates economic weakness, a fall in yields, and a larger number of buy orders in the futures market. The differing impact of macroeconomic news on order imbalances across the yield curve is also apparent when disaggregating into positive and negative news events. Bank bill futures react significantly to negative news on Building Approvals, CPI, GDP and Retail sales; the only significant reaction to positive news is with GDP and the effect on order imbalance is less than for negative news. Negative retail sales data is a significant determinant of order imbalance on both Bank bills and 3-year bonds. Order imbalance for both bond futures has a strong relationship to both positive and negative PPI news, with a stronger reaction to positive news. The only consistent relationship between a news event and order imbalance across all three futures contracts concerns negative CPI news; this is no surprise given the monetary policy stance and the keen focus on CPI taken by the RBA during the sample period.

<Insert Table 8>

The quasi-maximum likelihood estimates of the EGARCH model for the overall effect of scheduled macroeconomic news, as modelled in equations (7.A) and (7.B) are reported in Table 9. Significant and negative news coefficients are found in the conditional mean equations for all three futures, consistent with positive news surprises resulting in higher yields and thus falling futures prices. The impact of macroeconomic news is greater in the bond futures than on the bank bill futures.

In the conditional variance equation, the lagged variance term (β_h) is close to one in all cases, suggesting persistence of returns. The positive and significant results for (β_{e2}) the volume effects of innovations provide evidence that unexpected changes in the mean have the significant impact of raising conditional volatility. There is also significant evidence ($\beta_{e1} > 0$) that a rise in the conditional mean of returns leads to higher conditional variances for all considered futures. Order imbalance appears to have a positive significant effect on volatility for 90-day Bank Bills and bond futures, this increased sensitivity to order flow is possibly a result of an increase in the level of information asymmetry within the IR futures market. Positive news on building approvals,

employment, GDP, and wage costs tend to increase volatility, whilst positive news on CPI and Private CapEx tends to have a negative effect on conditional volatility. For retail sales the result is mixed with positive news resulting in rising volatility in bank bills, but falling volatility in 3- and 10-year bonds.

<Insert Table 9>

In summary, the release of scheduled major macroeconomic data has a significant impact on order imbalance, with order imbalance swiftly reacting to announcements, and in a manner that correctly reflects the news component. The pattern of order imbalance around announcements suggests that there is no information leakage prior to the announcement. The impact on bond futures of macroeconomic news is greater than for the shorter-term bank bill futures. Shorter-maturity futures react more closely to factors which have a bearing on near-term central bank action, whilst the longer-maturity bond futures are affected by inflation expectations. Following a macroeconomic announcement there is an increase in the level of information asymmetry within the Australian interest rate futures market demonstrated by an increase sensitivity to order flow.

5. Conclusion

The relations between trading activity and market returns have been explored extensively, particularly with regards the US equity markets. Trading activity has typically been measured by volume, but more recent work (e.g. Spiegel and Subrahmanyam (1995) and Chordia et al (2002)) suggests that the imbalance between buyer and seller initiated orders could be a powerful determinant of price movements beyond trading volume. Utilizing order imbalance methodology in the context of the Australian Interest Rate Futures market provides key findings which have implications for market participants and the implementation of trading strategies.

Order imbalances are strongly related to contemporaneous returns even after considering aggregate market activity. Contemporaneous order imbalance exerts a significant impact on market returns in the expected direction i.e. excess buy (sell) orders drive up (down) prices. However when predicting next period volatility aggregate volume disappears as a significant factor which suggests that order imbalance is a more informative measure.

Order imbalances are related to past market returns. Participants in the Australian Interest Rate futures market tend to be consistently contrarian in nature following market rallies, but whereas investors in bond futures continue to sell following market declines, bank-bill investors remain contrarian.

Scheduled macroeconomic data has a significant impact on order imbalance. Nine major macroeconomic announcements are identified. Order imbalance reacts swiftly to such announcements, and in a manner that correctly reflects the news component. The impact of macroeconomic news on bond futures is greater than that for the shorter-term bank bill futures. Longer-maturity bond futures are more strongly impacted by inflation news, whilst shorter-maturity futures react more robustly to news on economic growth.

There is an increase in information asymmetry following macroeconomic announcements. Following a scheduled macroeconomic announcement there is an increase in the level of information asymmetry within the interest rate futures market demonstrated by an increased sensitivity to order flow. However, the pattern of order imbalance prior to announcements suggests that there is no information leakage.

The relationship between trading activity and the return on financial assets provides a rich avenue for future work. Further work in this field could investigate the information leakages between different markets, both with markets of differing asset classes and on an international basis. In particular, the spill-over effects of macroeconomic announcements and the subsequent price discovery process are deserving of attention.

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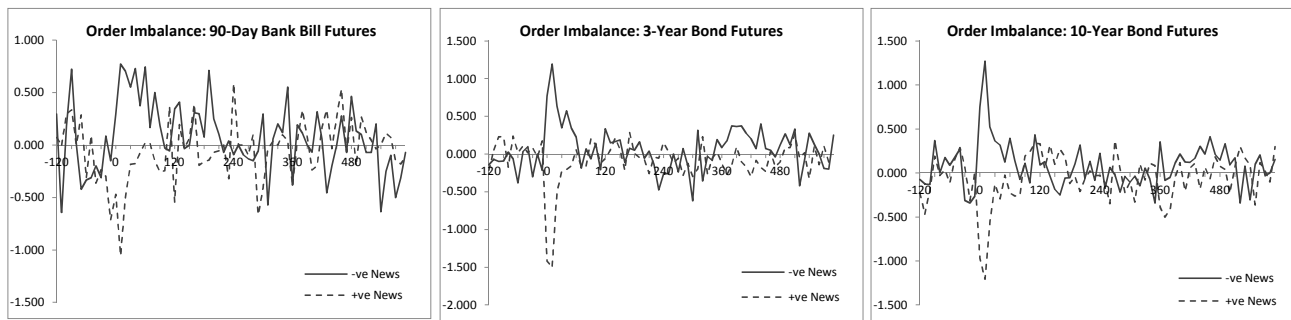


Figure 1. Order Imbalance around scheduled 11.30am macroeconomic announcements

Order imbalance patterns for each futures contract are plotted by 10-second intervals for days on which scheduled 11.30am macroeconomic announcements occur. The data is disaggregated by periods in which negative and positive news occurs. Times shown are deviations from the scheduled announcement time, in seconds. Data: Jan 2004 - Dec 2010

Table 1**Descriptive statistics and contract specifications for Australian interest rate futures**

This table reports descriptive statistics and contract specifications for the primary interest rate futures traded on the Sydney Futures Exchange (SFE). Volatility is measured as the daily price movement in the respective futures contract over the sample period. Volume is the average daily number of contracts traded over the same sample period. Turnover is measured in A\$ million. Note that the tick value is variable due to market pricing convention and moves in accordance with interest rates Data: Jan 2004 - Dec 2010

Contract	Reuters Code	Contract Size	Min Tick (%)	Value of 1 b.p.	Price Volatility	Ave. Daily Volume (Contracts)
90-Day Bank Bill Futures	YBA	A\$1,000,000	0.010	A\$24	0.0066	16,300.60
3-Year Government Bond Futures	YTT	A\$100,000	0.005	A\$28	0.0065	66,357.18
10-Year Government Bond Futures	YTC	A\$100,000	0.005	A\$38	0.0059	26,517.87

Table 2 **Summary statistics and correlations**

Descriptive statistics are provided for average daily order imbalance measures for Australian Interest Rate Futures; 90-Day Bank Bills, 3-year Government Bond Futures, 10-year Government Bond Futures. Trades are signed using the Lee and Ready (1991) algorithm. OIBTRA and OIBVOL measure the value-weighted order imbalance in number of transactions and contracts respectively. NUMTRANS is the average number of daily transactions. VOLUME is the average daily volume of contracts traded. Returns are measured as the average daily return. Values in bold face (Panel B and Panel C) are significantly non-zero with p-value less than 0.001. Data: Jan 2004 - Dec 2010

Panel A: Summary Statistics

90-Day Bank Bills					3-Year Bonds					10-Year Bonds				
	Mean	Median	Std. Dev.		Mean	Median	Std. Dev.			Mean	Median	Std. Dev.		
OIBTRA	-	2.35	-	1.00	42.92	3.71	3.00	91.63		8.20	3.00	156.39		
OIBVOL	151.96	49.00			4,801	1,019	544	9,537		276	136	3,213		
QSPR	0.01	0.01			0.04	0.01	0.01	0.03		0.01	0.01	0.00		
NUMTRANS	168.81	135.00			133.44	789.78	721.00	391.79		975.54	905.00	429.24		
VOLUME	16,301	13,754			12,473	66,357	60,769	34,806		26,518	23,997	13,841		
Returns (x10 ³)	0.0078	-			0.0066	0.0088	-	0.0065		0.0084	-	0.0059		

Panel B: Correlations

	OIBTRA	OIBVOL	QSPR	NUMTRANS	VOLUME		OIBTRA	OIBVOL	QSPR	NUMTRANS	VOLUME		OIBTRA	OIBVOL	QSPR	NUMTRANS	VOLUME
OIBVOL	0.398					OIBVOL	0.443					OIBVOL	0.438				
QSPR	-0.004	-0.016				QSPR	0.003	-0.006				QSPR	0.010	-0.016			
NUMTRANS	-0.058	0.018	0.023			NUMTRANS	0.198	0.012	-0.005			NUMTRANS	0.007	0.060	0.036		
VOLUME	0.055	0.070	0.015	0.580		VOLUME	0.037	0.089	-0.060	0.540		VOLUME	0.004	0.098	-0.266	0.541	
Returns	0.164	0.171	-0.015	0.103	0.013	Returns	0.317	0.426	-0.485	0.036	0.024	Returns	0.207	0.381	0.021	0.000	0.010

Panel C: Autocorrelations

Lag (days)	OIBTRA	OIBVOL	QSPR	VOLUME	Returns	Lag (days)	OIBTRA	OIBVOL	QSPR	VOLUME	Returns	Lag (days)	OIBTRA	OIBVOL	QSPR	VOLUME	Returns
1	1.511	1.697	2.001	1.206	1.850	1	1.668	1.819	2.002	1.264	2.037	1	1.593	1.709	1.161	0.903	2.070
2	1.675	1.820	2.002	1.474	1.930	2	1.584	1.876	2.002	1.502	1.984	2	1.678	1.856	1.318	1.143	2.044
3	1.729	1.956	2.002	1.604	1.936	3	1.712	1.869	2.003	1.584	1.938	3	1.794	1.893	1.308	1.183	1.983
4	1.762	1.919	2.003	1.644	2.047	4	1.744	1.945	2.003	1.522	2.012	4	1.959	2.036	1.363	1.096	1.980
5	1.758	1.898	2.005	1.698	2.018	5	1.724	2.012	2.003	1.483	1.963	5	1.917	2.016	1.313	1.054	1.980

Table 3 The impact of 11:30 macroeconomic announcements on volatility

The dependent variable in the regression is price volatility in the 30-second interval following announcements, interval j, on day t. Explanatory variables are the scheduled 11:30 (AEST) macroeconomic announcements. A significance level of 0.005 is used to identify major announcements. Data: Jan 2004 - Dec 2010

Announcement	90-Day Bank Bill			3-Year Bond			10-Year Bond		n	Frequency
	Coefficient	t-statistic		Coefficient	t-statistic		Coefficient	t-statistic		
Intercept	0.281	1.700	*	0.564	0.040		0.399	7.550	***	
ANZ Job Advertisements	0.698	1.170		0.923	0.020		0.450	1.360		84 Monthly
Average Weekly Wages	0.109	0.110		0.413	0.010		0.396	1.230		28 QoQ
Building Approvals	-0.566	-0.910		2.210	0.050		1.670	8.370	***	84 Monthly
Company Operating Profit	-0.490	-0.480		0.291	0.000		0.590	1.810	*	28 QoQ
Construction Work Done	-0.257	-0.270		0.572	0.010		-0.093	-0.290		28 QoQ
Consumer Price Index	5.770	5.800	***	61.696	8.140	***	7.570	23.800	***	28 QoQ
Current Account Balance	0.414	0.400		-0.374	0.000		0.294	0.890		28 QoQ
Dwelling Starts	0.696	0.700		0.369	0.000		0.233	0.730		28 QoQ
Employment	3.750	6.350	***	1.219	2.270	**	8.030	42.570	***	84 Monthly
Export price index	-0.160	-0.160		-0.166	0.000		-0.016	-0.050		29 QoQ
Gross Domestic Product	-2.330	-2.280	**	7.380	2.090	**	4.810	14.730	***	28 QoQ
Home Loans	-0.785	-0.670		0.982	0.010		0.907	1.410		84 Monthly
House Price Index	-0.098	-0.100		0.250	0.000		0.276	0.860		28 QoQ
Inventories	-0.490	-0.480		0.291	0.000		0.591	1.810		28 QoQ
Investment Lending	0.860	0.640		0.255	0.000		0.463	1.090		64 Monthly
Job vacancies	-0.291	-0.260		-0.150	0.000		0.036	0.110		22 QoQ
NAB Business Confidence	0.222	0.370		0.821	0.020		0.377	1.950	*	84 Monthly
New Motor Vehicle Sales	-0.120	-0.200		0.134	0.000		0.156	0.820		84 Monthly
Private Capital Expenditure	0.583	0.580		3.780	0.050		2.360	7.310	***	28 QoQ
Private Sector Credit	-0.006	-0.010		0.755	0.020		0.462	1.380		84 Monthly
Producer Price Index	2.450	2.460	**	5.740	2.080	**	3.300	10.360	***	28 QoQ
RBA Target Rate Decision	9.300	14.510	***	0.810	2.010	**	0.460	2.250	**	77 Monthly
Retail Sales	1.730	2.740	***	5.310	2.110	**	3.530	17.480	***	84 Monthly
Trade Balance	0.558	0.940		0.167	0.000		0.211	1.110		84 Monthly
Wage Cost Index	0.763	0.760		1.970	0.030		1.870	5.820	***	28 QoQ
Adjusted R ²	0.148			0.136			0.629			

The reported coefficients are 10³ times actual coefficients

Note: *, **, *** denotes significance at 10%, 5%, and 1%, respectively

Table 4 Causation of market order imbalance

Dependent variable is the order imbalance measured in number of transactions, OIBTRA_t, on trading period t. Explanatory variables are lagged values of OIBTRA_t, past positive and negative parts of market returns, and day of the week dummies. Data: Jan 2004 - Dec 2010

	90-Day Bank Bills		3-Year Bonds		10-Year Bonds	
	OIBTRA		OIBTRA		OIBTRA	
Intercept	-1.254		5.542		-17.038	*
	(-0.53)		(0.96)		(-1.66)	
OIBTRA _{t-1}	0.192	***	0.142	***	0.199	***
	(8.00)		(5.68)		(8.20)	
OIBTRA _{t-2}	0.093	***	0.159	***	0.131	***
	(3.83)		(6.34)		(5.30)	
OIBTRA _{t-3}	0.073	**	0.067	***	0.049	**
	(2.99)		(2.68)		(1.99)	
OIBTRA _{t-4}	0.071	**	0.053	**	-0.031	
	(2.99)		(2.15)		(-1.26)	
Min(0,R _{t-1})	10288.0	***	-8552.9	***	-39607.0	***
	(3.17)		(-3.42)		(-3.53)	
Min(0,R _{t-2})	7177.8	**	5097.8		-10054.0	
	(2.21)		(0.85)		(-0.90)	
Min(0,R _{t-3})	-96.7		11389.0	*	7237.1	
	(-0.03)		(1.91)		(0.65)	
Min(0,R _{t-4})	-11991.0	***	4814.5		1898.0	
	(-3.69)		(0.81)		(0.17)	
Max(0,R _{t-1})	-1642.4		-17868.0	**	-11472.0	
	(-0.90)		(-3.00)		(-1.02)	
Max(0,R _{t-2})	-3289.9	*	-706.8		-11895.0	
	(-1.81)		(-0.12)		(-1.06)	
Max(0,R _{t-3})	659.3		7425.0		8408.2	
	(0.36)		(1.24)		(0.75)	
Max(0,R _{t-4})	1199.0		5138.3		115.6	
	(0.65)		(0.85)		(0.01)	
Monday	-5.158		-5.569		18.119	
	(-1.65)		(-0.84)		(1.59)	
Tuesday	6.387	**	6.614	**	39.554	***
	(2.06)		(2.00)		(3.47)	
Wednesday	2.630		2.932		7.458	
	(0.85)		(0.44)		(0.65)	
Thursday	1.166		0.577		19.856	*
	(0.38)		(0.09)		(1.74)	
Durbin-Watson	2.030		2.010		2.010	
Adjusted R ²	0.105		0.089		0.085	

Note: *, **, *** denotes significance at 10%, 5%, and 1%, respectively

Table 5 Bond futures market returns, contemporaneous and lagged order imbalances and lagged returns

Panel A: The dependent variable is the return on the respective bond future. Explanatory variables include contemporaneous and lagged positive and negative order imbalances measured in number of trades (OIBTRA_t)/1,000 and lagged positive and negative market returns for respective futures markets. Panel B: A predictive regression if fit using observations that are common to the top 20% of days with high buy order imbalance as well as the top 20% of days with high returns. Another regression is run for observations on periods with high sell order imbalance and large negative returns. Data: Jan 2004 - Dec 2010

Panel A

Dependent Variable: R_t	90-Day Bank Bills				3-Year Bonds				10-Year Bonds			
Intercept	-0.386 *	-0.477 **	-0.314	-0.097	-0.280	-0.357	-0.194	0.168	0.200	0.201	0.007	0.004
	(-1.78)	(-2.14)	(-1.50)	(-0.55)	(-1.08)	(-1.28)	(-0.74)	(-0.08)	(1.02)	(0.85)	(0.03)	(0.18)
Excess Buy Orders	0.035 ***	0.033 ***			0.024 ***	0.024 ***			0.006 ***	0.006 ***		
	(5.29)	(5.08)			(8.63)	(8.55)			(4.10)	(4.03)		
Excess Sell Orders	0.022 ***	0.023 ***			0.022 ***	0.022 ***			0.013 ***	0.011 ***		
	(3.73)	(3.96)			(7.25)	(7.29)			(7.40)	(7.36)		
Lag(Excess Buy Orders)	0.010	0.007	0.011 *		0.0002	-0.0003	0.0000		-0.001	-0.001	0.0004	
	(1.50)	(1.02)	(1.68)		(0.06)	(-0.12)	(1.53)		(-0.75)	(-0.68)	(0.26)	
Lag(Excess Sell Orders)	-0.016 ***	-0.016 ***	-0.007		-0.006 **	-0.006 *	0.0000		-0.003 **	-0.003 **	-0.0004	
	(-2.73)	(-2.67)	(-1.29)		(-2.01)	(-1.89)	(-0.49)		(-2.12)	(-1.99)	(-0.26)	
Lag(Positive return)		0.100 ***	0.100 ***	0.108 ***		0.032	-0.012	0.004		-0.013	-0.025	-0.024
		(3.52)	(3.48)	(3.80)		(0.78)	(-0.29)	(0.09)		(-0.30)	(-0.59)	(-0.57)
Lag(Negative return)		-0.023	-0.007	-0.163		-0.017	-0.041	-0.042		-0.014	-0.045	-0.047
		(-0.45)	(-0.14)	(-0.32)		(-0.41)	(-0.95)	(-0.99)		(-0.33)	(-1.04)	(-1.10)
Adjusted R^2	0.034	0.041	0.010	0.008	0.103	0.104	0.002	0.001	0.049	0.046	0.001	0.001
No. Observations	1795	1795	1795	1795	1795	1795	1795	1795	1795	1795	1795	1795

Panel B

Dependent Variable: R_{t+1}	90-Day Bank Bills				3-Year Bonds				10-Year Bonds			
	Days with OIBTRA in top quintile and R_t in top quintile		Days with OIBTRA in bottom quintile and R_t in bottom quintile		Days with OIBTRA in top quintile and R_t in top quintile		Days with OIBTRA in bottom quintile and R_t in bottom quintile		Days with OIBTRA in top quintile and R_t in top quintile		Days with OIBTRA in bottom quintile and R_t in bottom quintile	
	top quintile	bottom quintile	top quintile	bottom quintile	top quintile	bottom quintile	top quintile	bottom quintile	top quintile	bottom quintile	top quintile	bottom quintile
Intercept	0.620 ***	0.050 ***	-0.604 ***	-0.050 ***	0.959 ***	0.001 ***	-0.945 ***	-0.836 ***	0.886 ***	1.020 ***	-0.866 ***	-0.817 ***
	(9.83)	(4.41)	(-12.45)	(-6.05)	(24.62)	(12.32)	(-19.73)	(-8.24)	(24.05)	(13.25)	(-20.59)	(-9.12)
Lagged order imbalance (OIBTRA _t)	0.006	0.004	-0.004	-0.003	0.009 **	0.009 **	-0.004 **	-0.004 **	0.0065	0.005	-0.004	-0.005
	(0.49)	(0.36)	(-0.43)	(-0.30)	(2.11)	(2.12)	(-2.23)	(-2.39)	(0.03)	(0.21)	(-1.57)	(-1.64)
Lagged return (R_t)	0.114	0.100	-0.024	-0.263	0.097 *	0.104 *	-0.201 ***	-0.194 ***	-0.063	-0.070	-0.229	-0.020
	(1.91)	(1.64)	(-0.46)	(-0.51)	(1.77)	(1.91)	(-2.82)	(-2.73)	(-1.23)	(-1.36)	(-0.37)	(-0.32)
Lagged Volume (Volume _t)		0.000		0.000		0.000 *		0.000		0.000 *		0.000
		(1.30)		(-1.54)		(-1.71)		(-1.22)		(-1.94)		(-0.62)
Adjusted R^2	0.027	0.038	0.0043	0.022	0.039	0.059	0.061	0.072	0.0133	0.044	0.024	0.027
No. Observations	146	146	132	132	144	144	138	138	121	121	121	121

Note: *, **, *** denotes significance at 10%, 5%, and 1%, respectively

Table 6 Absolute market returns, order imbalance, volume and liquidity

The dependent variable is the absolute value of the daily return in each respective futures market. Explanatory variables include contemporaneous and lagged positive and negative order imbalances measured in number of trades ($OIBTRA_t$) / 1,000, aggregate volume / 1×10^6 and lagged returns $|R_{t-1}|$. Data: Jan 2004 - Dec 2010

	90-Day Bank Bills			3-Year Bonds			10-Year Bonds		
	$ R_t $	$ R_{t+1} $		$ R_t $	$ R_{t+1} $		$ R_t $	$ R_{t+1} $	
Intercept	-0.190 (-0.81)	0.166 *** (6.77)		0.132 *** (5.37)	0.424 *** (17.58)		-0.787 *** (-7.28)	-0.335 *** (-2.99)	
Excess buy orders	0.018 *** (3.11)	0.015 ** (2.48)		0.009 *** (5.14)	0.005 *** (2.69)		0.004 *** (4.34)	0.004 *** (4.30)	
Max(0, $OIBTRA_t$)									
Excess sell orders	0.012 ** (2.45)	0.014 *** (2.73)		0.001 *** (5.56)	0.009 *** (4.21)		0.003 *** (2.60)	0.001 (1.05)	
$ \text{Min}(0, OIBTRA_t) $									
Volume _t	0.001 *** (12.53)	0.002 * (1.77)		2.00E-05 *** (9.59)	-1.40E-05 * (-1.80)		0.001 *** (8.12)	-8.65E-07 (-1.23)	
Quoted Spread _t	0.069 (0.22)	-0.018 (-0.05)		0.171 (0.52)	-0.129 (-0.38)		0.186 *** (9.80)	0.133 *** (6.78)	
$ R_{t-1} $	0.106 *** (4.68)	0.133 *** (5.38)		0.164 *** (7.32)	0.203 *** (8.43)		0.094 *** (4.08)	0.112 *** (4.65)	
Adjusted R ²	0.122	0.034		0.117	0.058		0.102	0.060	

Note: *, **, *** denotes significance at 10%, 5%, and 1%, respectively

Table 7 Dynamics of order imbalance around scheduled 11:30am macroeconomic data releases

This table demonstrates the dynamics of order imbalance around scheduled 11:30am macroeconomic data releases. For each interest rate future the order imbalance is shown for the period 11:29:30 to 11:31:00. Data is aggregated across all days (including those without data releases) and then disaggregated into days with at least one scheduled releases and no scheduled releases. Finally the days on which scheduled releases occur are disaggregated into negative news events and positive news events. Data: Jan 2004 - Dec 2010

OIBTRA	(-30,-20)	(-20,-10)	(-10,0)	(0,10)	(10,20)	(20,30)	(30,40)	(40,50)	(50,60)	(60,70)	(70,80)	(80,90)	(90,100)	(100,110)	(110,120)
90-Day Bank Bills															
Aggregate	-0.168	-0.021	-0.279	-0.115	-0.034	0.107	0.107	0.184	0.100	0.208	0.168	0.158	0.084	-0.204	0.090
No News	-0.113	-0.237	-0.212	-0.115	0.079	0.117	0.018	0.117	0.052	0.047	0.261	0.163	0.184	-0.260	0.000
News	-0.273	-0.156	-0.382	-0.115	-0.103	0.101	0.180	0.248	0.144	0.355	0.082	0.151	-0.013	-0.147	0.164
-ve News	-0.313	0.083	-0.150	0.295	0.771	0.702	0.551	0.727	0.373	0.744	0.167	0.500	0.186	-0.029	-0.057
+ve News	-0.235	-0.300	-0.714	-0.471	-1.057	-0.500	-0.185	-0.177	-0.075	0.020	0.018	-0.152	-0.257	-0.244	0.368
3-Year Bonds															
Aggregate	-0.046	-0.011	-0.101	-0.218	-0.100	0.101	0.028	0.033	0.067	-0.024	-0.061	0.040	0.095	0.149	-0.043
No News	-0.027	-0.006	-0.142	-0.058	0.007	0.156	-0.007	-0.108	0.042	-0.175	0.038	0.124	0.109	0.165	0.044
News	-0.080	-0.019	-0.021	-0.368	-0.199	0.045	0.063	0.177	0.091	0.135	-0.175	-0.052	0.078	0.132	-0.162
-ve News	-0.302	0.000	-0.217	0.767	1.195	0.637	0.349	0.575	0.343	0.224	-0.186	0.070	-0.061	0.139	-0.206
+ve News	0.088	-0.043	0.174	-1.417	-1.503	-0.507	-0.230	-0.201	-0.157	0.054	-0.165	-0.162	0.203	0.125	-0.122
10-Year Bonds															
Aggregate	0.025	-0.152	-0.021	-0.058	0.064	-0.026	0.052	0.050	0.017	0.031	-0.027	0.011	0.081	0.091	0.265
No News	0.067	-0.008	0.021	-0.022	0.140	-0.004	-0.004	0.098	-0.004	0.004	0.015	0.149	0.017	0.102	0.176
News	-0.096	-0.329	-0.141	-0.112	-0.019	-0.051	0.113	-0.013	0.043	0.067	-0.079	-0.177	0.153	0.076	0.385
-ve News	-0.313	-0.341	-0.267	0.750	1.269	0.525	0.364	0.318	0.124	0.394	0.144	-0.073	0.111	-0.111	0.433
+ve News	0.073	-0.317	0.000	-0.966	-1.206	-0.574	-0.126	-0.295	-0.024	-0.229	-0.261	0.255	0.194	0.242	0.337

Table 8 Order Imbalance and the effect of macroeconomic news

This table depicts regression output for equation (5.A) and (5.B). The dependent variable is the order imbalance, measured in number of trades (OIBTRA_t) during period t.

Explanatory variables are lagged order-imbalance and news effects. The 1st column for each contract contains the output for the aggregate news effect, the 2nd and 3rd columns for the news effect disaggregated into positive and negative news. Data: Jan 2004 - Dec 2010

Dependent Variable: OIBTRA _t	90-Day Interbank						3-Year Bond			10-Year Bond		
	EQ 5.A:	EQ 5.B:		EQ 5.B:	EQ 5.A:	EQ 5.B:	EQ 5.B:	EQ 5.A:	EQ 5.B:	EQ 5.B:		
	All news	+ve news	-ve news	All news	+ve news	-ve news	All news	+ve news	-ve news			
Intercept	0.011 (0.75)	0.007 (0.38)			-0.007 (-0.61)	-0.016 (-1.28)		0.019 ** (1.97)	0.020 * (1.87)			
OIBTRA _{t-1}	0.122 *** (11.19)	0.121 *** (11.09)			0.117 *** (18.52)	0.116 *** (18.37)		0.124 *** (19.73)	0.123 *** (19.63)			
Building Approvals	-0.154 *** (-3.56)	-0.134 * (-1.79)	0.162 *** (2.88)		-0.142 *** (-4.02)	-0.212 *** (-3.65)	0.085 * (1.80)	-0.063 * (-1.93)	-0.098 * (-1.92)	0.035 (0.80)		
Consumer Price Index	-0.118 ** (-2.46)	-0.105 (-1.38)	0.128 ** (2.02)		-0.184 *** (-3.85)	0.008 (0.11)	0.326 *** (5.13)	-0.120 *** (-2.73)	-0.016 (-0.24)	0.198 *** (3.37)		
Employment	-0.065 ** (-1.99)	-0.049 (-1.42)	0.224 * (1.94)		-0.078 *** (-3.11)	-0.054 (-1.93)	0.178 *** (2.83)	-0.101 *** (-4.33)	-0.111 *** (-4.26)	0.052 (0.87)		
Gross Domestic Product	-0.301 *** (-4.42)	-0.238 ** (-2.45)	0.365 *** (3.76)		-0.110 ** (-2.16)	-0.129 (-1.90)	0.082 (1.05)	0.046 (0.93)	0.033 (0.51)	-0.062 (-0.82)		
Private Capital Exp.	0.026 (0.28)	-0.162 (-1.20)	-0.208 (-1.56)		-0.009 (-0.15)	0.008 (0.10)	0.027 (0.29)	0.001 (0.01)	0.049 (0.69)	0.051 (0.69)		
Producer Price Index	-0.070 (-1.12)	-0.098 (-1.20)	0.027 (0.28)		-0.312 *** (-5.48)	-0.353 *** (-4.98)	0.229 ** (2.35)	-0.254 *** (-4.65)	-0.360 *** (-5.15)	0.084 *** (2.95)		
Retail Sales	-0.145 *** (-3.55)	-0.124 * (-1.81)	0.158 *** (2.93)		-0.084 *** (-2.86)	-0.019 (-0.36)	0.121 *** (3.20)	-0.011 (-0.39)	0.012 (0.23)	0.022 (0.60)		
Wage Cost Index	-0.141 (-1.55)	-0.161 (-0.95)	0.135 (1.25)		-0.054 (-0.77)	-0.054 (-0.39)	0.058 (0.71)	-0.110 * (-1.70)	-0.063 (-0.44)	0.122 (1.67)		

Note: *, **, *** denotes significance at 10%, 5%, and 1%, respectively

Table 9 EGARCH(1,1) estimations of returns in 12-minute period around scheduled 11:30 macroeconomic data releases

This table reports the quasi-maximum likelihood estimates of the EGARCH(1,1) model, as described in equation (7.A) and (7.B), of returns in Australian interest rate futures in the interval around scheduled macroeconomic announcements. Data: Jan 2004 - Dec 2010

Panel A: Conditional Mean Equation				Panel B: Conditional Variance Equation					
	90-day Bank Bill	3-Year Bond	10-Year Bond		90-day Bank Bill	3-Year Bond	10-Year Bond		
α_c	0.0003 (1.10)	0.0003 (0.98)	0.0003 (0.98)	β_c	-15.5146 (-64.82)	*** (-13.94)	-0.3766 (-13.96)	***	-0.3899 (-13.96)
$\alpha_{lag,1}$	0.0283 (1.16)	-0.0033 (-0.12)	0.0054 (0.18)	β_h	0.9945 (27.04)	*** (73.12)	0.9623 (58.17)	***	0.9611 (58.17)
				β_{e1}	0.0586 (4.38)	*** (6.80)	0.0495 (6.71)	***	0.0509 (6.71)
				β_{e2}	0.0711 (3.78)	*** (10.61)	0.0704 (10.87)	***	0.0719 (10.87)
$\alpha_{Ordersign}$	8.67E-06 (1.74)	* (1.75)	5.76E-06 (1.75)	* (1.75)	$\beta_{Ordersign}$	0.0013 (3.33)	*** (8.73)	***	0.0035 (8.66)
$\alpha_{Building App.}$	-0.0038 (-3.98)	*** (-6.42)	-0.0072 (-6.45)	*** (-6.45)	$\beta_{Building App.}$	-0.0470 (-0.52)	0.1244 (4.08)	*** (4.04)	0.1254 (4.04)
α_{CPI}	-0.0250 (-31.71)	*** (-59.75)	-0.0443 (-60.19)	*** (-60.19)	β_{CPI}	-0.0914 (-2.01)	** (-4.35)	*** (-4.05)	-0.1045 (-4.05)
$\alpha_{Employment}$	-0.0206 (-41.48)	*** (-54.18)	-0.0352 (-53.98)	*** (-53.98)	$\beta_{Employment}$	0.6314 (15.16)	*** (13.70)	*** (13.69)	0.2456 (13.69)
α_{GDP}	-0.0112 (-9.02)	*** (-12.28)	-0.0214 (-12.23)	*** (-12.23)	β_{GDP}	-0.0695 (-0.67)	0.1880 (4.42)	*** (4.23)	0.1811 (4.23)
$\alpha_{Private CapEx}$	-0.0055 (-4.54)	*** (-2.43)	-0.0041 (-2.41)	** (-2.41)	$\beta_{Private CapEx}$	0.0345 (0.34)	-0.2180 (-4.32)	*** (-4.26)	-0.2194 (-4.26)
α_{PPJ}	-0.0080 (-7.90)	*** (-7.69)	-0.0124 (-7.73)	*** (-7.73)	β_{PPJ}	-0.3545 (-7.55)	*** (5.39)	*** (5.10)	0.1461 (5.10)
$\alpha_{Retail Sales}$	-0.0085 (-16.80)	*** (-20.57)	-0.0133 (-19.87)	*** (-19.87)	$\beta_{Retail Sales}$	0.1173 (3.66)	*** (-5.43)	*** (-5.47)	-0.1169 (-5.47)
$\alpha_{Wage Cost}$	-0.0034 (-1.61)	-0.0061 (-6.44)	*** (-6.50)	*** (-6.50)	$\beta_{Wage Cost}$	-0.1159 (-0.42)	0.7055 (11.05)	*** (10.95)	0.7087 (10.95)
α_{Mon}	-0.0008 (-0.85)	0.0007 (-0.12)	-0.0005 (-0.71)		β_{Mon}	0.0418 (1.94)	* (1.67)	* (1.91)	0.0869 (1.91)
Adjusted R ²	0.310	0.385	0.385						
Log Likelihood	5773.78	5323.31	5323.24						
Durbin Watson	2.046	1.969	1.970						

Note: *, **, *** denotes significance at 10%, 5%, and 1%, respectively